

The Future of Protein Scaffolds as Affinity Reagents for Purification

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ABSTRACT: Affinity purification is one of the most powerful separation techniques extensively employed both at laboratory and production scales. While antibodies still represent the gold standard affinity reagents, others derived from non-immunoglobulin scaffolds emerged as interesting alternatives in particular for affinity purification. The lower costs of production, fast ligand development, and high robustness are appealing advantages of non-immunoglobulin scaffolds. These have successfully been used in the affinity purification of relevant targets as antibodies, human serum albumin, transferrin, and other biomarkers, as reviewed in this work. Furthermore, a critical assessment on the strengths, weaknesses, opportunities, and threats related with the implementation of non-immunoglobulin scaffolds as ligands in affinity purification are discussed.

Biotechnol. Bioeng. 2017;114: 481–491.

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KEYWORDS: affinity purification; protein scaffolds; affinity ligands; non-immunoglobulin; biopharmaceuticals; downstream processing

Introduction

Protein engineering has revolutionized various fields, from enzyme improvement for technical applications to new therapeutic tools for human diseases. Within the area of affinity purification, protein engineering has also been critical to develop new affinity ligands, namely antibodies and antibody-based architectures for new targets (Cuesta et al., 2010), or to generate improved versions of natural biological ligands (Güllich et al., 2002). In the biopharmaceutical industry, engineered versions of protein A derived from *Staphylococcus aureus* (SpA) are currently the standard affinity ligands used in the capture step of mAb purification. However, the available affinity resins for laboratory or industrial applications are still expensive and account for a significant proportion of the total production costs (60–80%) (Gronemeyer et al., 2014). Several efforts are in place to amplify the portfolio of affinity ligands for purification, in an attempt to decrease costs and increase selectivity (Gottschalk et al., 2012). Furthermore, the increasing number and variety of biological molecules and particles entering the market, adds an extra pressure in the required purification processes.

The design and selection of affinity ligands for purification typically addresses issues as the introduction of suitable chemistry for immobilization in purification matrices, the stability to harsh chemical conditions (e.g., organic solvents or extreme pH) and to “cleaning-in-place” or “sterilization in place” procedures, the selectivity and binding capacities over multiple cycles. A plethora of affinity ligands with distinct scaffolds, ranging from biological to small synthetic ligands, has been explored. These possess fairly different sizes and levels of complexity, being the most simple and less costly small synthetic ligands and pseudobispecific ligands (Fang and Zhang, 2008) (Fig. 1, Table I). For example, protein based ligands tend to be less stable during harsh elution and cleaning cycles, leading to leakage and, consequently, a decrease in the capacity and lifetime of the resin (Hedhammar et al., 2005).

Natural protein binders are expensive to produce, have low resistance to chemical immobilization strategies, to elution and regeneration conditions, and are prone to protease attack (Binz and Pluckthun, 2005; Taussig et al., 2007). For example, antibodies employed in immunoprecipitation, are complex glycosylated proteins composed of two different polypeptide chains with disulfide bonds. Antibody production is performed through

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The authors declare no conflicts of interest.

Abbreviations: aa, amino acids; ABD, albumin binding domain; CBM, carbohydrate binding module; CD, circular dichroism; CIP, cleaning in place; CSF, cerebrospinal fluid; CV, column volumes in AKTA systems; DSC, differential scanning calorimetry; *E. coli*, *Escherichia coli*; ELISA, enzyme-linked immunosorbent assay; EDC, 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride; HEK 293T cells, human embryonic kidney cells; Her 2, Her 2 receptor; HRP, horseradish peroxidase; HSA, human serum albumin; Ig, immunoglobulin; IgG, immunoglobulin G; mAbs, monoclonal antibodies; non-Ig, non-immunoglobulin; NHS, N-hydroxysuccinimide; Ni-NTA, nickel-nitrilotriacetic acid; NTPs, nucleotide triphosphate; SpA, *Staphylococcus aureus* protein A; SPPS, solid-phase peptide synthesis; SPR, surface plasmon resonance; T_m, temperature of melting; TNF- α , tumor necrosis factor; WT, wild type; PulD, membrane secretin bacterial protein; HEWL, hen egg white lysozyme from chicken; IMAC, immobilized-metal affinity chromatography; Sulfo-NHS, N-hydroxysulfosuccinimide.

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Contract grant sponsor: Fundação para a Ciência e Tecnologia

Contract grant number: UID/Multi/04378/2013

Contract grant sponsor: ERDF

Contract grant number: POCI-01-0145-FEDER-007728

Contract grant sponsor: Fundação para a Ciência e Tecnologia (FCT)

Contract grant number: SFRH/BD/72664/2010

Received 14 July 2016; Revision received 25 August 2016; Accepted 26 August 2016

Accepted manuscript online 26 August 2016;

Article first published online 26 September 2016 in Wiley Online Library
(<http://onlinelibrary.wiley.com/doi/10.1002/bit.26090/abstract>).

DOI 10.1002/bit.26090

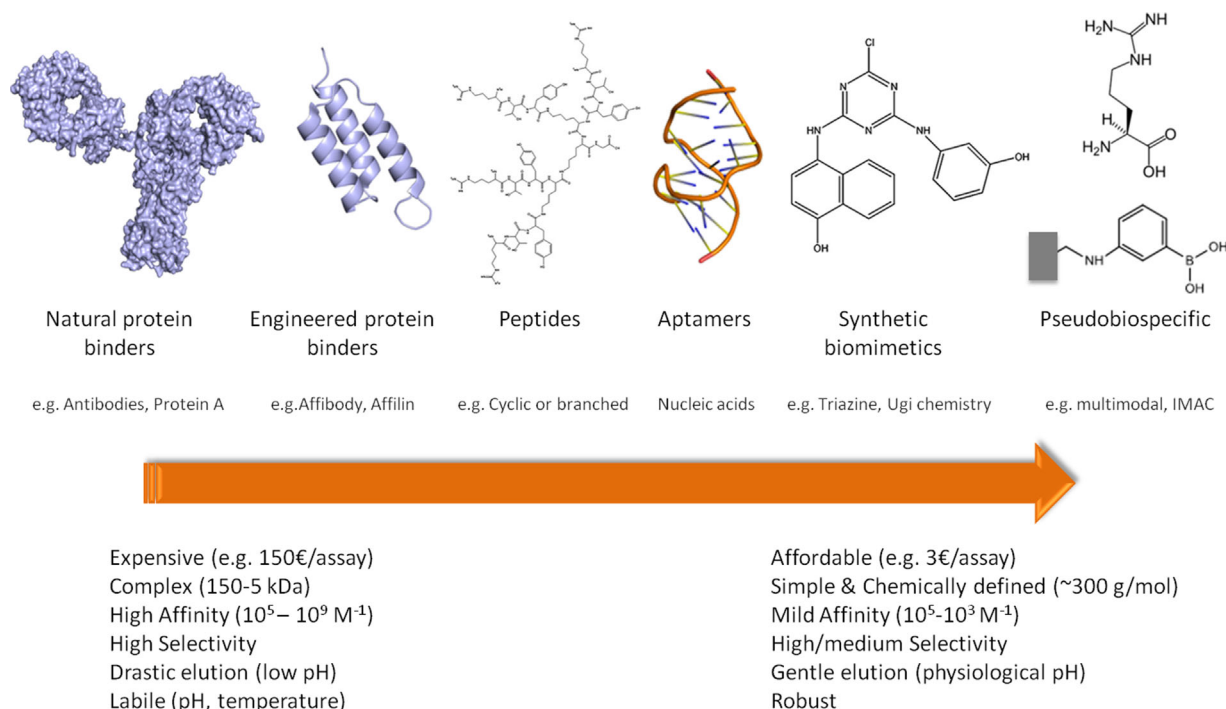


Figure 1. Affinity reagents properties used in bioseparation. The affinity reagents have very different size and consequently different complexities of production and costs. The synthetic ligands have lower specificity; however, have high stability under temperature and pH conditions. In contrast, antibodies are highly specific for their targets, but are less stable under harsh conditions.

Table I. Reagents for affinity purification.

Name	Affinity reagent description
Natural protein binders	Biological ligands identified in natural systems. These include antibodies for a certain target, lectins binding to carbohydrate moieties, or protein A from SpA binding to antibodies. The interactions between the natural binders and the target biomolecules are usually very strong ($K_a \sim 10^6 - 10^8 \text{ M}^{-1}$). Natural biospecific ligands are widely employed both at laboratorial and industrial-scale processes (Colwill and Graslund, 2011; Shukla and Thömmes, 2010).
Engineered protein binders	Protein scaffolds derived from natural systems, which were evolved through yeast, phage display or cell free display or rationally designed against a target. The most described protein scaffolds are the Affibody, Adnectin, DARPin, Monobody, Knottin, Affitins, Kunitz domain, PDZ-domain or the Lipocalins. There are recent extensive reviews on each protein scaffold (Binz et al., 2005; Gebauer and Skerra, 2009; Mouratou et al., 2015), as well as on their applications in biomedicine (e.g., therapy and diagnostic) and structural biology (Škrlec et al., 2015). More recently, promising protein scaffolds, such as Sso7d (Béhar et al., 2016; Gera et al., 2011, 2012) or T7 phage gene 2 protein (Gp2) (Kruziki et al., 2015), both proteins were evolved to target proteins, peptides, but also a small organic molecule.
Peptides	Peptides with different lengths and geometries (e.g., linear, cyclic (Fassina et al., 2001) or branched (Ruvo et al., 2011)) have been developed and optimized, some with purification performances compared to equivalent full-proteins (Naik et al., 2011). These peptides are usually identified by screening combinatorial libraries of peptides (Heldt et al., 2009; Yang et al., 2009) obtained by chemical synthesis or biological evolution (Kelley et al., 2004; Mooney et al., 2011), and further coupled to separation matrices.
Aptamers	Aptamers are synthetic nucleic acid-based affinity ligands, isolated from combinatorial libraries of synthetic nucleic acids by an iterative process of selection and amplification termed SELEX (systematic evolution of ligands by exponential enrichment) (Javaherian et al., 2009; Kanoatov et al., 2010; Kokpinar et al., 2011). These are not very popular and applied in restricted cases.
Synthetic biomimetics	Engineered ligands designed for high selectivity toward the target molecule. A common feature of engineered ligands is the existence of a scaffold that is decorated with various functionalities. Fully synthetic reagents include designed Triazine-based ligands (Roque et al., 2007) or Ugi-based ligands (Haigh et al., 2009). Both result from a combination of rational design, solid-phase combinatorial synthesis and screening of restricted libraries. The affinity constants of the interaction between the lead ligands and the target proteins are usually lower and with minor contact surface areas when compared to natural biological ligands (Pina et al., 2014). Apart from some successful examples, they tend to lack specificity and the degree of variability is more restricted. However, they are cheaper to produce and extremely resistant to biological and chemical degradation.
Pseudobiospecific	Within the group of pseudobiospecific ligands, we can find small ligands (e.g., thiophilic, metal chelating), which are quite general and lack selectivity for specific targets (Batalha et al., 2012; Pina et al., 2014), for example imac, amino acid ligands (Sousa et al., 2009, 2010, 2011).

expression in eukaryotic cells systems, and purification mostly relies on resins with protein A, which increases their production cost (Skerra, 2007). Antibodies are less stable in very harsh differences in pH, use of organic solvents, and high salt concentrations can lead to denaturation (Wörn and Plückthun, 2001). This can be problematic specifically for purification purposes where antibodies are immobilized in a chromatographic matrix and can leach from the support thus contaminating the final product (Hedhammar et al., 2005). However, antibodies have high specificity against their targets, since they are naturally evolved to be affinity proteins. They have large interaction surfaces, thus can promote several types of interactions, for example, hydrogen bonds or hydrophobic interactions in the same contact region.

In recent years, engineered protein scaffolds became interesting to solve some of the problems observed in the antibodies, such as the lack in stability (e.g., chemically or thermal) while maintaining the specificity and have low costs of production (Boersma and Plückthun, 2011). These proteins can be isolated from different origins, such as microorganisms, plants, or humans. They have been used in biomedical applications—as part of therapeutic drug conjugates, protein fusions, or diagnostic tools (Škrlec et al., 2015); and, for biotechnological applications—as affinity reagents for purification or in protein microarrays (Béhar et al., 2016). They can be used in the same applications as natural protein binders and can have an improved performance in some applications due to their stability.

To simultaneously overcome the limitations of fully synthetic ligands and natural biological binders, engineered protein domains with the stability of synthetic ligands and the specificity of antibodies represent interesting alternatives.

Protein Domains as Affinity Ligands

Affinity reagents derived from protein domains include peptide sequences derived from Immunoglobulins (Ig) fragments and non-Immunoglobulins (non-Ig) structures. In general, they have higher selectivity than synthetic affinity ligands, as they present a higher surface of interaction, and they can be further selected to adapt to a specific recognition region in the target protein (Fig. 1). Another advantage include the high stability and the easy and less costly production by *Escherichia coli* expression or by chemical synthesis (Binz et al., 2005; Sawyer et al., 2014). These structures can be further optimized to improve their stability for different pHs, temperature, or chemicals, through the mutation of critical amino acids (Stoevesandt and Taussig, 2007). Furthermore, protein engineering can be used to generate combinatorial libraries of Ig fragments or non-Ig structures, with partially or totally randomized regions, which are evolved against different targets. This is possible through yeast, phage, ribosome, or cell free display systems (Grönwall and Ståhl, 2009). An ideal protein scaffold to develop affinity reagents for purification processes must be a very robust, small scaffold (less than 20 KDa) (Uhlén and Hober, 2009), with well-defined regions for diversity and stability (highly conserved regions that maintain folding of the protein and other variable regions, which when totally or partially randomized do not affect the stability of the overall protein); possess a peptide (or peptide like) recognition area to facilitate interaction with biological

molecules; have no cysteines and disulfide bridges, and no need for post-translational modifications; be easy to synthesize by biological and chemical means; be straightforward to randomize by biological and chemical evolution mechanisms; be resilient toward the introduction of un-natural components; allow the easy, oriented, and stable conjugation to solid matrices; and possess stability toward conditions present in the final application. Furthermore, modifications must not be limited to loops (like in antibodies), as other flat surfaces or rigid loops can be changed. Therefore, scaffolds can be selected to adapt to the contact surface in the target, allowing a better interaction between the scaffold and the target protein reducing unspecific binding (Ponsel et al., 2011; Sawyer et al., 2014).

Combinatorial libraries based on different protein domains became increasingly powerful tools to generate affinity reagents against different targets. The libraries were firstly tested with Ig fragments, where modifications in non-conserved regions were introduced through total or partial randomization. The introduction of mutations can be a combination of the 20 amino acids or can be guided by structural or functional information of the target protein, where a rational design approach is followed. In both strategies, the aim is to increase or gain a specific function but maintain the structure, which usually represents a major challenge. Libraries based in a protein sequence are constructed through manipulation of genetic material, with the incorporation of mutations via degenerated codons. The libraries are selected against a target, and after several rounds of selection, it is possible to identify protein binders (Grönwall and Ståhl, 2009). These strategies allowed to engineer antibody fragments and also generate single domains (e.g., diabodies, minibodies, or V-domains) against different targets (Holliger and Hudson, 2005). The success of antibody engineering was limited by the lack of stability. The presence of cysteine residues limits the fusion with other proteins, since the fusion can be prone to aggregation. In addition, antibodies are often associated with a high cost of production (Binz et al., 2005; Rothe et al., 2006). Therefore, the search for alternative non-Ig structures that could be more versatile, easy to produce, and less costly became appealing.

The different families of protein scaffolds are commonly divided in two categories: globular proteins and repeats proteins. Globular proteins include domains that are formed by only one peptide chain (e.g., Affibody (Fig. 2) (Löfblom et al., 2010)) and the repeat proteins chains have two or more repeats of one domain (e.g., designed ankyrin repeat proteins (DARPs) (Plückthun, 2015)) or the Armadillo Repeat Proteins ((Fig. 2) (Sawyer et al., 2014)). Despite the diversity of protein scaffolds developed by several academic research groups and companies, only some have been exploited as affinity ligands for purification processes, which are the focus of the present review.

Globular Proteins

Affibody

Affibodies are derived from the Z domain, a modified structure of the B domain of SpA (Fig. 2B). This is a three-helix structure with 58 residues (6 KDa), without disulphide bonds, and independent folding (Braisted and Wells, 1996) that is easily expressed and chemically produced through different strategies

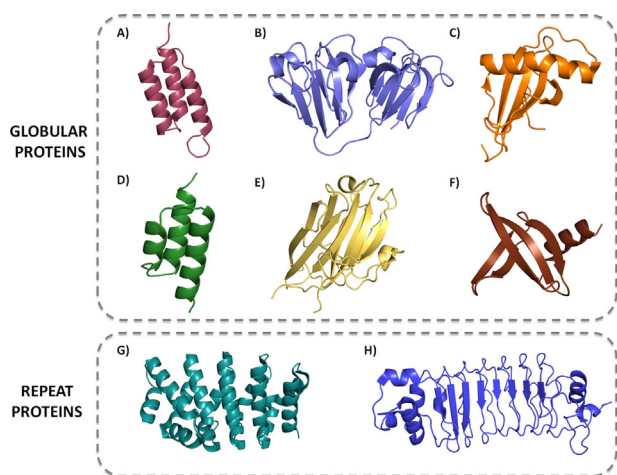


Figure 2. Protein scaffolds structures used in bioseparation. Globular proteins: (A) Affibody (PDB code: 1Q2N); (B) Affilin (PDB code: 2JDG); (C) Affimer (PDB code: 1NB5, chain I); (D) Albumin Binding Domain (ABD) from Streptococcal Protein G (PDB code: 1GJT, 18–65 residues); (E) Carbohydrate Binding Module (CBM) (PDB code: 1K45); (F) Nanofitin (PDB code: 4CJ2, chain C). Repeat proteins: (G) Armadillo Repeat Proteins (PDB code: 4DB6); (H) Repebody (PDB code: 4J4L, chain A).

(Lindgren et al., 2012). This structure was extensively studied and used for evolution using phage and ribosome display against different targets. The randomized regions are typically localized in the α -helix region. This is the contact surface with the targets in the natural protein A, and the number of randomized residues is commonly 13 (Nygren, 2008). A biotech company Affibody AB is currently exploiting the Affibody molecules as scaffolds for the identification of therapeutic solutions for different diseases. The Affibody scaffold has been engineered to target human receptors important in different diseases—cancer, the breast cancer marker ErbB2 (Her2) with a 22pM affinity (Wikman et al., 2004), or rheumatoid arthritis, TNF- α (Löfdahl et al., 2009). Affibodies were also explored for sensing applications—diagnostic; immobilized in microarrays to capture proteins; or, as a fusion partner of other proteins (Justino et al., 2015; Nygren, 2008). The binding of Taq DNA polymerase to the affibody generated the common HOT Start DNA polymerases. In this particular case, the enzyme is inhibited to bind DNA at low temperatures, due to the interaction with the affibody, and is only active at higher temperatures (98°C). It was demonstrated that the affibody could be immobilized to allow the purification of the enzyme, with high purity and binding in the micromolar range. The support could be reused after clean-up with alkaline solution (0.5 M NaOH). In the same conditions, it was possible to bind and elute the Human Apolipoprotein A-1 (Nord et al., 2000). In addition, an anti-protein A Affibody was selected with a nanomolar range affinity, which allows to capture SpA fusions and dimers with a picomolar range affinity (Lindborg et al., 2013) (Table II). But other target proteins have been purified by Affibody-based resins, as Transferrin (Grönwall et al., 2007b), or Alzheimer amyloid β -peptides (Grönwall et al., 2007a) (Table III). Due to their potential, they were explored to bind

other proteins, namely for depletion of HSA and Transferrin from human serum plasma or blood (Fang and Zhang, 2008). Recently, a novel support based on bacterial cellulose modified with an anti-HSA affibody (immobilized through carbodiimide crosslinker chemistry—EDC/NHS covalent coupling) was able to capture 81 ng of HSA/cm² (Orelma et al., 2014). For the Transferrin purification, a phage display selected affibody was tested with samples from human plasma, and from human cerebrospinal fluid (CSF), whereas specificity was assessed with HSA and IgG. Affibodies targeting Transferrin and antibodies anti-HSA and anti-IgG were immobilized in cross-linked agarose through a Cys residue. The Transferrin resin was tested and the recovered samples were passed through resins containing anti-HSA and anti-IgG antibodies. Finally, a serum sample depleted from Transferrin, HSA, and IgG was obtained. This was also valid for human CSF. Consequently, it was possible to remove up to 68% of plasma Transferrin in one cycle and two cycles were used to improve the depletion. For human CSF, it is possible to deplete 78% of Transferrin in a single cycle (Grönwall et al., 2007b). More recently, with the purpose of depleting high abundant proteins in CSF, several affibodies selected to bind HSA, IgG, Transferrin, and Transthyretin, as well as polyclonal antibodies against Cystatin C were immobilized in resins used to deplete the proteins in CSF samples. These resins were used consecutively to deplete the five proteins in a proportion of 99%, 95%, 74%, 92%, and 83% for HSA, IgG, Transferrin, Transthyretin, and Cystatin C, respectively. This strategy allowed the characterization of samples for the presence of low abundant proteins through mass spectrometry (Ramstrom et al., 2009).

Affibodies were also evolved to capture Alzheimer Amyloid β -peptides. The affibody was immobilized in a solid-support through a His-tag, since the selected binders had a Cys residue in the target recognition region. This support was tested with samples from plasma and serum spiked with the peptides, showing high specificity and an affinity of 230–320 nM toward amyloid peptides. This strategy can be useful to capture these peptides in blood and decrease Amyloid plaques formation (Grönwall et al., 2007a).

As an application note, the Affibody AB company indicates the procedure for purification of insulin from human serum. The affibody anti-insulin with a free SH-group from a Cys was immobilized in a resin with an iodoacetyl group (Sulfo-Link coupling, Pierce), a common methodology described by (Jonsson et al., 2008), yields and affinity resin with a capacity of 1.8 mg human insulin/mL Anti-Insulin Affibody of support (Affibody) (Table III).

Affibodies have also been developed to act as tags for purification of recombinant fusion proteins in a positive or negative ion exchange column (Hedhammar et al., 2004; Hedhammar and Hober, 2007).

Several affibodies are commercially available immobilized already in a resin or ready for immobilization through a Cys residue, such as: Anti-EGFR Affibody, Anti-IgE Affibody, Anti-TNF- α Affibody, or Anti-Interleukin (IL-8), can be purchased through Affibody AB or Abcam, UK.

Affibodies have been applied in liquid chromatography columns for depletion of highly abundant proteins and are commercialized in

Table II. Protein scaffolds structures derived from non-Ig fragments and without Cys residues. Comparison between production and biophysical properties.

Name	Scaffold	Size (KDa)	Structure folding	Production	<i>T_m</i>	Ref.
Affibody	Z domain derived from B domain of SpA	6.5 (58 aa)	3 helix independent folding	<i>E. coli</i> expression (soluble form) Chemical synthesis and native chemical ligation	<i>Z_{WT}</i> 78°C Anti- <i>Z_{Taq}</i> and <i>Z_{Taq}</i> 55 and 57°C Two Z domains with S-S bound <i>T_m</i> increased 10-fold	(Lindgren et al., 2012; Nygren, 2008)
Affilin	Human g-crystallin/human ubiquitin	21 (176 aa)	β-sheet	<i>E. coli</i> expression (200 mg/L culture and 100 mg/L after purification) Yeast expression	56 and 80°C	(Fiedler et al., 2006)
Affimer	Human protease inhibitor stefin A	10 (98 aa)	4 β-sheet and 1 α-helix	<i>E. coli</i> and Mammalian cells expression	80°C	(Stadler et al., 2011)
Albumin binding domain	Albumin binding domain from protein G	5 (46 aa)	3 helix independent folding	<i>E. coli</i> expression	ABD _{WT} 70°C ABD 48°C (CD) 58°C (SPR)	(Jonsson et al., 2008; Linhult et al., 2002)
Carbohydrate binding module	Xylanase from <i>Rhodothermus marinus</i>	18 (170)	Several β-sheets	<i>E. coli</i> expression (40–100 mg/L)	CBM _{WT} 89°C	(Gunnarsson et al., 2004)
Nanofitins	Sac7d protein from <i>Sulfolobus acidocaldarius</i>	7 (66 aa)	4 β-sheets, 1 α-helix, and small loops	<i>E. coli</i> expression (high solubility)	Mutants 70–80 °C Sac7d _{WT} 89°C Sac7 mutants 66–89.6°C (DSC)	(Mouratou et al., 2007)
Armadillo repeat proteins	Armadillo β-catenin/rationally designed	18–27 (2 or 4 internal repeats) (42 aa, monomer)	3 α-helix	<i>E. coli</i> expression (100 mg/L)	70–85°C (CD)	(Parmeggiani et al., 2008; Varadamsetty et al., 2012)
Repebody	Variable lymphocyte receptors (VLRs) and leucine-rich repeat (LRR)	30 (20–30 aa repeats) (3 KDa per repeat)	β-strand-turn-α helix	<i>E. coli</i> expression (highly soluble, 50–80 mg/L)	70–85°C dependent on 3–6 repeats	(Lee et al., 2012)

WT, wild type protein; *T_m*, temperature of melting; CD, circular dichroism; SPR, surface plasmon resonance; DSC, differential scanning calorimetry.

pre-packed column format MARS-7 and MARS-14 columns from Agilent Technologies (Table IV).

Affilin

Affilin is derived from human g-B-crystallin, a protein found in eye lens in vertebrates (Fig. 2C). This is a structure with 176 aa that fold into several β-sheets, and is highly stable—for example, the bovine derived protein is stable in a wide range of pH (1–9), at high temperatures (75°C), and even in strong denaturation conditions (7M urea). These properties make it interesting for therapeutic applications, but especially for affinity purification purposes (Table II). The human g-B-crystallin structure was randomized by phage display at the N-terminal and in eight amino acid positions, with solvent-exposed side chains in β-sheets, against recombinant pro-form of nerve growth factor (proNGF). The characterization of the affilin for purification of this target was demonstrated. The anti-proNGF affilin was immobilized in a epoxy-activated polymethacrylate resin and the novel support is highly robust to regeneration conditions (stands pH range 1.6–12.5) and maintains its binding to

the target after these treatments (Fiedler et al., 2006) (Table III). Affilins were selected to bind the Fc fragment of IgG (10^{−8}M, 10 nM), estradiol and testosterone (100–200 nM) (Ebersbach et al., 2007), and also to extradomain B of fibronectin, a target almost exclusively expressed in tumor tissues (30 and 200 pM) (Lorey et al., 2014).

This scaffold is explored by SCIL Proteins GmbH for different applications as therapeutics, but also for capture of biomarkers for diagnostic using different solid-supports, spacer arms, and ligand formats (mono- or multivalent scaffolds) (Table IV).

Affimer

The Affimer scaffold is derived from a protease inhibitor Stefin A (cystatin A). The interest in this scaffold arises from its small size (10 KDa), high thermal (80°C) and chemical (pH 2–12) stability, besides the conformation with two loops that can be engineered in 12–36 positions (Hoffmann et al., 2010; Stadler et al., 2011) (Fig. 2D, Table II). Affimer-derived binders can be used for therapeutic applications, sensing (ELISA or Western-blot),

Table III. Protein scaffolds application in bioseparation.

Name	Selection	Library size	Targets	Affinity	Support used	Elution/binding capacity	Cycles of use	CIP stability	Ref.
Affibody	Phage display	10^7 – 10^9	Taq DNA polymerase	Micromolar range	—	—	—	Alkaline conditions using pulses of 0.5 M NaOH	(Nord et al., 2000)
			Human apolipoprotein A1						
		10^9	Human β -amyloid peptides	230–320 nM (SPR)	HiTrap affinity columns (NHS-activated Sepharose) HP and Streptavidin HP columns	0.3 M HAc, pH 2.8	Potentially recycled	Regeneration using elution buffer	(Grönwall et al., 2007a)
		10^9	Transferrin	400 nM	N-terminal Cys immobilized in sulfo-link coupling gel	0.2 M Glycine-HCl 0.1 M NaCl and 1 M urea, pH 2.2	—	—	(Grönwall et al., 2007b)
	n.d.	n.d.	Human serum albumin	10 nM Affibody (#ab31897, MW 14 kDa)	Bacterial cellulose by EDC/NHS-mediated covalent conjugation and physical adsorption	81 ng/cm ² support	—	—	(Orelma et al., 2014)
	n.d.	n.d.	Several proteins from human cerebrospinal fluid (HSA, IgG, Transferrin and Transthyretin)	—	N-terminal Cys immobilized in Sulfo-Link coupling gel	0.5 M HAc	—	—	(Ramstrom et al., 2009)
Affilin	Phage display	10^8	Recombinant pro-form of nerve growth factor (proNGF)	0.006 μ M	Epoxy-activated polymethacrylate resin and	1 M urea 0.5 M HAc, pH 2.8	10 cycles	25 mM HCl, 6 M Urea, 6 M GdmHCl, 20% ethanol, 5 M NaCl or 10 mM NaOH in this conditions is stable up to 10 cycles	(Ebersbach et al., 2007; Fiedler et al., 2006)
						1.8 mg/mL support 0.1 M glycine pH 2.2			
Affimer	Phage display	10^{10}	Complement C3b (CC3B3)	—	His-tagged Affimer was immobilized in Ni-NTA beads	—	—	4 h at 0.5 M or 0.1 NaOH or 0.1 M Glycine pH2 or 18 h at pH13	^a
Albumin binding domain	Phage Display	10^9	Human serum albumin	1st generation 3.8–5 pM	N-terminal Cys immobilized in Sulfo-Link coupling gel	0.5 M HAc pH 2.8	—	—	(Jonsson et al., 2008)
				2nd generation 50–500 fM (WT = 1.2 nM)		Theoretical capacities of 2 mg/column (AKTA system)			
Carbohydrate binding module	Phage display	10^6	Sacharides	120 μ M	Silica 5–50 mg protein/mL silica gel bed volume	Operation at 25–65°C	160 min at 65°C decrease the capacity in 7%	—	(Gunnarsson et al., 2004; Johansson et al., 2006)

(Continued)

TABLE III. (Continued)

Name	Selection	Library size	Targets	Affinity	Support used	Elution/binding capacity	Cycles of use	CIP stability	Ref.
Nanofitins	Ribosome display	10 ¹²	hIgG, bacterial PulD-N protein and chicken HEWL protein	—	Affitins anti-hIgG, anti-PulD-N or anti-HEWL were immobilized in HiTrap NHS-activated column	Elution: glycine buffer pH 2.5	25 cycles	0.25 M NaOH for 15 min up to a maximum of 6h15 min decrease in binding capacity between 10–26% for anti-IgG support	(Béhar et al., 2016)
			hIgG lysozyme protein	9.3 × 10 ⁵ M ⁻¹	Magnetic nanoparticles	100 mM glycine-HCl buffer, 0.15 M NaCl, pH 2.5	—	Stable in different buffers: pH ranging from 2.5 to 11 50% ethylene glycol	(Fernandes et al., 2016)
						Binding capacity: 10 mg IgG, 16 mg PulD and 10.8 mg HEWL per mL of column packaging with purity between >92%.			
Armadillo repeat proteins	Ribosome display	10 ¹¹	Peptide neurotensin	5.0 × 10 ⁴ M ⁻¹ 7 μM	—	—	—	Denaturation using 4 M GndHCl	(Varadamsetty et al., 2012)
Repebody	Phage display	—	Fc fragment of human IgG	0.19 mM	NHS-activated sepharose 4 fast flow resin	10 mM sodium acetate pH 4.0	10 cycles	0.5 M NaOH for 15 days	(Heu et al., 2014)
						Purity >94.6% Recovery >95.7%			

n.d., no data; HAc, acetic acid; SPR, surface plasmon resonance; CIP stability, clean in place stability; ^o, online case study for affinity purification; HEWL, hen egg-white lysozyme.

diagnostic and also for affinity purification (Straw et al., 2013), and are being exploited by the Avacta Life Sciences company.

One example of application in affinity purification is the enrichment of Complement C3b (CC3B3) from human serum. The enriched sample was characterized through mass spectrometry and clearly the protein of interest was present in higher amount in comparison with other contaminants (Table III). The affimer is selected against the target and modified to have a Cys residue or a His-tag, to facilitate the immobilization in solid-supports using different chemistries (e.g., His-tag can be captured by Ni-NTA beads) (Table IV).

Albumin-Binding Domain

The albumin-binding domain (ABD) derived from Streptococcal protein G is a small domain of 46 amino acids with a three helix folding. ABD was affinity matured by phage display to increase affinity to Human Serum Albumin (HSA) from a nanomolar to

femtomolar range (Fig. 2A). The protein was engineered in 15 out of 46 amino acids, located in the contact region between the ABD and HSA. The variants identified after in vitro selection presented improved affinity; however, some positions in the protein were conserved when compared with the wild-type. Therefore, further rational design was used to produce a second generation of ligands (Jonsson et al., 2008). One of the variants with improved properties was applied in HSA purification (Jonsson et al., 2008), and also in half-life extension of therapeutic proteins by generation of bivalent domains that bind HSA and Tumor Necrosis Factor-α (TNF-α) (Nilvebrant and Hober, 2013). The HSA-binder was produced with a Cys residue to promote its immobilization on cross-linked agarose for HSA capture from human serum. The binding capacity was estimated at 2 mg HSA/1 mL of Sulfo-Link gel (Pierce), and the protein was recovered using an acidic elution buffer (0.5 M acetic acid pH 2.8). The affinity constant for this system was estimated by surface plasmon resonance to be in the femtomolar range (Jonsson et al., 2008) (Table II and III).

Table IV. Protein scaffolds commercially available for bioseparation.

Name	Commercial name	Solid support immobilization	Kits	Targets	Recovery	Ref.
Affibody	Affibody	Cross-linked agarose	Multiple affinity removal column human (MARS14, 6 or 7)	Albumin, IgG, antitrypsin, IgA, transferrin, haptoglobin, fibrinogen, alpha2-macroglobulin, alpha1-acid glycoprotein, IgM, apolipoprotein AI, apolipoprotein AII, complement C3 and transthyretin or Mouse albumin, IgG, and transferrin	MARS14 94%	Affibody, Sweden http://www.affibody.com/
			Mouse 3 Several Affibodies Customizable	Mouse albumin, IgG, and transferrin Insulin or TNF- α Biomarkers capture for diagnostic, such as exosomes	MARS6 and 7 88–92% — —	Agilent Technologies, USA http://www.agilent.com/ Abcam, UK SCIL Proteins http://www.scilproteins.com/
Affilin	—	Resins, magnetic beads or membranes	Customizable	—	—	Avacta Life Sciences https://www.avactalifesciences.com/
Affimer	—	Resin beads	Customizable	—	—	Affilogic, France http://www.affilogic.com/
Nanofitins	—	Surfaces, membranes or resins	Under development through Merck Millipore	Vaccines (glycoconjugates, protein antigens or viruses)	—	Affilogic, France http://www.affilogic.com/

Carbohydrate Binding Module

The carbohydrate binding module (CBM) is a small domain (18 KDa) derived from a xylanase from *Rhodothermus marinus* Xyn 10 A (CBM 4–2 protein) (Fig. 2E, Table II). This structure is thermally very stable and binds to soluble poly- or oligosaccharides of xylose and glucose, thus it has been classified as a type B binding module (Guillén et al., 2010). It has been evolved, through randomization of 12 residues using a phage display library of 10⁶ clones, against different targets including xylan cellulose, mannan, and a human glycoprotein (a monoclonal IgG4 antibody). The selections for the saccharides were performed at different temperatures 25 and 60°C (Gunnarsson et al., 2004). This scaffold was employed in weak affinity chromatographic separation of saccharides using selected CBM immobilized in silica, employing an isocratic gradient of temperatures between 25 and 65°C. The capacity of the support was maintained after treatments at 65°C during 160 h (Johansson et al., 2006) (Table III). Thus, this work can indicate a possible application for purification of other targets after immobilization of CBM in solid-supports.

Nanofitins

Nanofitins (also designated as Affitins) were derived from protein Sac7d, a dsDNA-binding protein from *Sulfolobus acidocaldarius* (Fig. 2F). This is a protein of 66 residues (7 KDa), without disulphide bonds, and since it is derived from a thermophile organism, it is highly thermally and chemically stable, even at extreme pH (Edmondson and Shriver, 2001) (Table II). Due to its origin, nanofitins are easily expressed in *E.coli*, consequently making its production fairly easy and inexpensive (Béhar et al., 2013).

This protein was randomized at the surface in two short loops using ribosome display against different targets: Lysozyme, for

inhibition of glycosidase activity (Correa et al., 2014), secretin PulD protein (Mouratou et al., 2007), or human IgG. In the later, 14 residues were randomized and selected. It was observed that binding of clone C3 (from L1 library) to IgG was inhibited by SpA, suggesting that the epitope recognized is on the Fc domain of IgG. Another selected clone (D1, Library L2) was not inhibited by SpA to bind IgG, therefore, indicating another binding site on Fc (Béhar et al., 2013).

Nanofitins are being explored by the company Affilogic (Table IV), for applications as anti-TNF- α therapeutics for treatment of inflammatory bowel disease, through oral administration, due to the high stability of these proteins versus antibodies.

Nanofitins can be immobilized onto matrices through His-tags, phosphorylated serine residues, and Cys residues. These strategies allow an oriented immobilization onto different surfaces (e.g., glass slides, resins, or magnetic supports), thus extending the potential application of nanofitins. Nanofitins can be used for detection in microarrays (Cinier et al., 2009, 2012; Miranda et al., 2011), or affinity purification using as a solid-support chromatographic resins or membranes (Béhar et al., 2016) (Table IV). Recently, the use of different nanofitins for the purification of IgG, secretin PulD protein and hen egg white lysozyme (HEWL) from chicken was demonstrated. The supports were able to capture and elute all proteins (purity higher 92%), and even after 25 cycles of resin use and CIP conditions with 0.25 M NaOH, binding capacity higher was than 76% (Béhar et al., 2016). In addition, anti-lysozyme and anti-IgG affitins were immobilized through a Cys using Sulfo-SMCC chemistry in a magnetic support, which was used for the capture of lysozyme and human IgG from *E.coli* supernatant and human plasma, respectively. The supports were able to capture 58 mg lysozyme/g support and 165 mg IgG/g support and after purification both proteins were obtained in high purity >95%. Furthermore, the affinity constant for anti-lysozyme and anti-IgG supports was determined by static binding studies to be

$5.0 \times 10^4 \text{ M}^{-1}$ and $9.3 \times 10^5 \text{ M}^{-1}$, respectively (Fernandes et al., 2016). The potential for bioseparation will be further explored in purification of vaccines: glycoconjugates, protein antigens, and viruses, now supported by the European project DiViNe (<http://divineproject.eu/>).

Repeat Proteins

Armadillo Repeat Proteins

The armadillo repeat proteins family is encoded by the *Armadillo* locus, a DNA region important in *Drosophila* embryogenesis. The stability of the structures were optimized computationally by modifying the N and C-terminal with repeats to protect the hydrophobic core from solvent exposure (Parmeggiani et al., 2008) (Fig. 2G). The resulting structures demonstrated high thermal stability, and are commonly used at neutral to high pH (Alfarano et al., 2012) (Table II).

The native structures have been used in pull-down studies to capture families of proteins. In one example, armadillo repeats were expressed with FLAG tag and immobilized in anti-FLAG antibody resins. The modified resins were incubated with mammalian crude extracts from HEK293T cells. After precipitation, the samples were analyzed by mass spectrometry. It was observed that the armadillo proteins could capture Striatin, SG2NA, and Zinedin, a small group of eukaryotic proteins called the Striatin family, which are mainly neuronal proteins (Breitman et al., 2008). Another example, a designed armadillo repeat protein was randomized in six positions in each repeat (42 aa) generating a diversity per repeat of 10^6 , and selected through ribosome display against peptide neurotensin. Neurotensin was immobilized in streptavidin beads and incubated with an *E. coli* extract containing the armadillo repeat, showing a very specific binding, despite the low affinity (range of $7 \mu\text{M}$) (Varadamsetty et al., 2012) (Table III).

Repebody

The Repebody is a scaffold based on variable lymphocyte receptors (VLRs), which consist in an assembly of three to six repeating Leucine-rich repeat (LRR) modules. These modules possess 20–29 residues and a β -strand-turn- α -helix structure (Fig. 2H, Table II). The repebody molecules demonstrated a high soluble expression in *E. coli*, and display high thermal (70 – 85°C) and pH (3 – 12) stability. Due to their modular structure, repebodies have been engineered to generate binders with distinct affinity against a target (Lee et al., 2014). For example, repebodies were evolved against interleukin-6 (IL-6), and lysozyme through phage display (Lee et al., 2012).

More recently, this modular domain was used for the purification of IgG. A repebody structure with six repeats was mutated in two modules, which were evolved using phage display against the Fc region of IgG. The structure was immobilized in an N-hydroxysuccinimide (NHS)-activated resin and the purification of IgG from CHO cells culture was achieved at low pH. This strategy allowed the recovery of more than 96% of the protein of interest with high purity (95%). This support was also very stable to regeneration and allowed reutilization (Heu et al., 2014) (Table III).



Figure 3. Identification of strengths, weaknesses, opportunities, and threats for the implementation of protein scaffolds as affinity ligands in downstream processing.

Concluding Remarks and Future Perspectives

Affinity reagents based on protein scaffolds derived from non-Ig fragments have become increasingly important. These protein scaffolds are developed to have high stability, with a cost-efficient production by expression in prokaryotic systems or by chemical synthesis; and finally, fit structurally in the target molecule, which increase the application success. Protein scaffolds from different families have been validated for several targets and applications. The affibody scaffold is the most implemented scaffold for affinity purification purposes, but other protein scaffolds possess the same potential use. Thus, we expect that in the upcoming years, the affibody success story can be translated to other protein scaffolds and implemented in lab-scale or industry purification processes (Fig. 3).

This work was supported by the Unidade de Ciências Biomoleculares Aplicadas-UCIBIO which is financed by national funds from FCT/MEC (UID/Multi/04378/2013) and co-financed by the ERDF under the PT2020 Partnership Agreement (POCI-01-0145-FEDER-007728). A.M.G.C.D. is grateful to Fundação para a Ciência e Tecnologia (FCT)—Portugal for funding through PhD grant SFRH/BD/72664/2010.

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